

Sustainability aware asset management

Lecture 10: ESG Investing

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Objectives of the Lecture

We now focus on the second main topic of sustainable finance: ESG investing

1. What are the main objectives of ESG investing?

- Investor's preference ("Doing good while doing well")
- Risk management ("Climate risk is investment risk", Larry Fink, 2021)
- Long-term performance

2. Main approaches

- ESG integration, exclusion, or best-in-class
- Engagement
- Impact investing

3. Main steps of the lecture

- We describe the market for ESG investing
- We illustrate the main strategies of ESG investing
- We show how it works in practice (CRML website)

- **Berg, F., J. Koelbel, and R. Rigobon** (2022): *Aggregate Confusion: The Divergence of ESG Ratings*, *Review of Finance*, 26 (6), 1315–1344
- **Alessandrini, F., and E. Jondeau** (2020): *ESG Investing: From Sin Stocks to Smart Beta*, *Journal of Portfolio Management*, Ethical Investing 2020, 46 (3), 75–94
- **Alessandrini, F., D. Baptista Balula, and E. Jondeau** (2022): *ESG Screening in the Fixed-Income Universe*, *Journal of Investment Management*, 20(4), 65–86

Objectives of the Lecture

- **Concepts and Definitions**
- Market of ESG Investing
- ESG Scoring
- ESG Strategies

Three concepts with a similar meaning

Sustainable investing

Sustainable investing is an investment approach that considers environmental, social and governance (ESG) factors in portfolio selection and management

Socially responsible investing (SRI)

Socially responsible investing (SRI) is an investment strategy that is considered socially responsible, because it invests in companies that have ethical practices

Environmental, Social and Governance (ESG)

Environmental, Social, and Corporate Governance (ESG) refers to the factors that measure the sustainability of an investment

7 ESG Strategies

1 Exclusion

2. Values

3. Selection

4. Thematic

5. Integration

6. Engagement

7. Impact

Negative/exclusionary screening

Norms-based screening

Best-in-class/positive screening

Sustainability themed/thematic investing

ESG integration

**Corporate engagement and
shareholder action**

**Impact investing and
community investing**

7 ESG Strategies

1. Negative/exclusionary screening

Exclusion from a fund or portfolio of certain sectors, companies, countries or other issuers based on activities considered not investable. Exclusion criteria can refer to product categories (e.g., weapons, tobacco), company practices (e.g., animal testing, violation of human rights, corruption) or controversies

2. Norms-based screening

Screening of investments against minimum standards of business or issuer practice based on international norms such as those issued by the UN, ILO, OECD, and NGOs

In Europe, the top exclusion criteria are (1) controversial weapons (Ottawa and Oslo treaties), (2) tobacco, (3) all weapons, (4) gambling, (5) pornography, (6) nuclear energy, (7) alcohol, (8) genetically modified organisms and (9) animal testing (Eurosif, 2019)

3. Best-in-class/positive screening

Investment in sectors, companies or projects selected for positive ESG performance relative to industry peers, and that achieve a rating above a defined threshold.

4. Sustainability themed/thematic investing

Investing in themes or assets specifically contributing to sustainable solutions – environmental and social (e.g., sustainable agriculture, green buildings, lower carbon tilted portfolio, gender equity, diversity).

5. ESG integration

Systematic and explicit inclusion by asset managers of ESG factors into financial analysis

7 ESG Strategies

6. Corporate engagement & shareholder action Employing shareholder power to influence corporate behavior, through direct corporate engagement (i.e., communicating with senior management and/or boards of companies), filing or co-filing shareholder proposals, and proxy voting guided by comprehensive ESG guidelines

7. Impact investing and community investing

Impact investing: Investing to achieve positive, social, and environmental impacts – requires measuring and reporting against these impacts, demonstrating the intentionality of investor and underlying asset/investee, and demonstrating the investor contribution

Community investing: Where capital is specifically directed to traditionally underserved individuals or communities, as well as financing that is provided to businesses with a clear social or environmental purpose. Some community investing is impact investing, but community investing is broader and considers other forms of investing and targeted lending activities.

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- Concepts and Definitions
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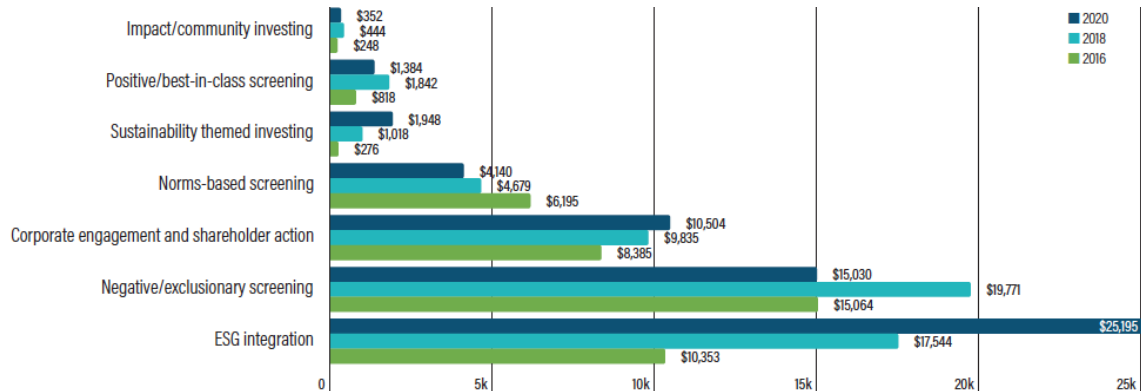
Global sustainable investing assets (USD billions)

REGIONS	2016	2018	2020
Total AUM of regions	81,948	91,828	98,416
Total sustainable investments only AUM	22,872	30,683	35,301
% Sustainable investments	27.9%	33.4%	35.9%
Increase of % sustainable investments (compared to prior period)		5.5%	2.5%

Global assets under management (USD billions)

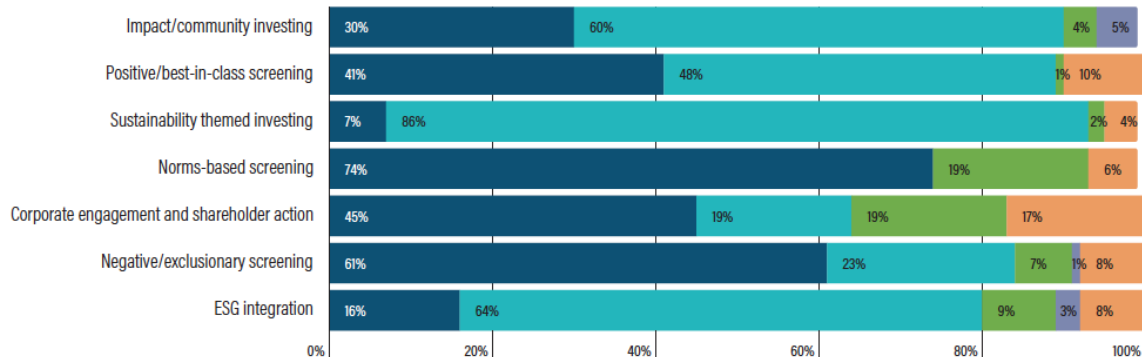
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Global growth of sustainable investing strategies 2016–2020



Source: GSIA (2021), Global Sustainable Investment Review

Regional share in global use of sustainable investing strategies 2016–2020



Source: GSIA (2021), Global Sustainable Investment Review

What Happened in 2022?

Numbers have dramatically changed in the 2022 survey.

US Sustainable Investment Forum changed its methodology: it does not include the AUM of investors who stated that they practice firmwide ESG integration but did not provide information on any specific ESG criteria they used in their investment decision-making

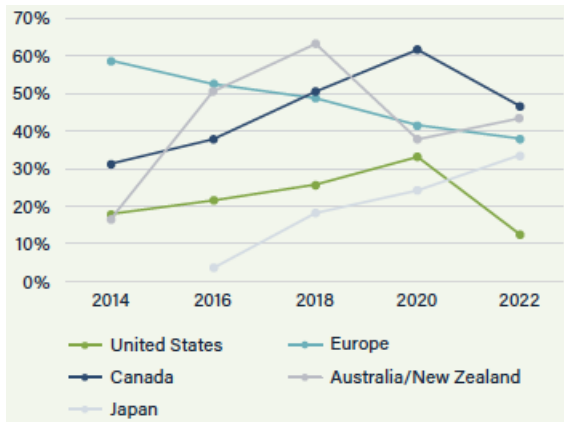
Several U.S. asset managers in the 2022 Trends Information Request reported far lower sustainable investing AUM than they had in 2020

REGION	2016	2018	2020	2022
Europe	12,040	14,075	12,017	14,054
Canada	1,086	1,699	2,423	2,358
Australia & New Zealand	516	734	906	1,220
Japan	474	2,180	2,874	4,289
Sub-total (USD Billions)	14,115	18,688	18,220	21,921
<i>% change</i>		32%	-3%	20%
United States	8,723	11,995	17,081	8,400
Total (USD Billions)	22,838	30,683	35,301	30,321
<i>% change</i>		34%	15%	n/a

Source: GSIA (2022), Global Sustainable Investment Review

What Happened in 2022?

Proportion of sustainable investing relative to total managed assets



REGION	2014	2016	2018	2020	2022
Europe	59%	53%	49%	42%	38%
United States	18%	22%	26%	33%	13%
Canada	31%	38%	51%	62%	47%
Australia & New Zealand	17%	51%	63%	38%	43%
Japan		3%	18%	24%	34%

Source: GSIA (2022), Global Sustainable Investment Review

Objectives of the Lecture

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Concepts of ESG Investing

- **ESG analysis**: extra-financial analysis $\neq$ financial analysis
- **ESG scoring**: quantitative measures of ESG dimensions
- **ESG ratings**: provide a grade (e.g., AAA, AA, A) to an issuer ($\approx$ credit ratings)
- **ESG screening**: process of scanning and filtering issuers based on ESG analysis and scoring ($\approx$ stock screening, bond screening, stock picking)
- **ESG investment process**: define how the investment process integrates ESG
- **ESG reporting**: provide ESG information and measures on the investment portfolio (e.g., ESG risk of the portfolio vs ESG risk of the benchmark, repartition of ESG ratings, top/bottom ESG issuers)

Some ESG Rating Agencies

Fundamental data providers: Broad range of publicly available raw data, from company reports or company websites

- Refinitiv
- Bloomberg

Comprehensive data providers: Combination of publicly available data from media, NGOs and company reports, and own proprietary questionnaires

- MSCI-ESG
- LSEG-Refinitiv-ESG
- Sustainalytics
- ISS ESG
- Vigeo-Eiris
- RobecoSAM
- TruValue Lab (daily update)
- RepRisk (daily update)

Specialized data providers: In-depth and highly contextualized data covering one or two ESG aspects (e.g., carbon emissions)

- CDP (Carbon Disclosure Project)
- TruCost (S&P Global)

LSEG-Refinitiv-ESG Scoring Methodology

LSEG-Refinitiv ESG (aka Datastream or Thomson Asset4) is the only data provider that discloses the **underlying data points** – numeric or Boolean values

Current coverage: > 10 000 listed firms in advanced and emerging markets

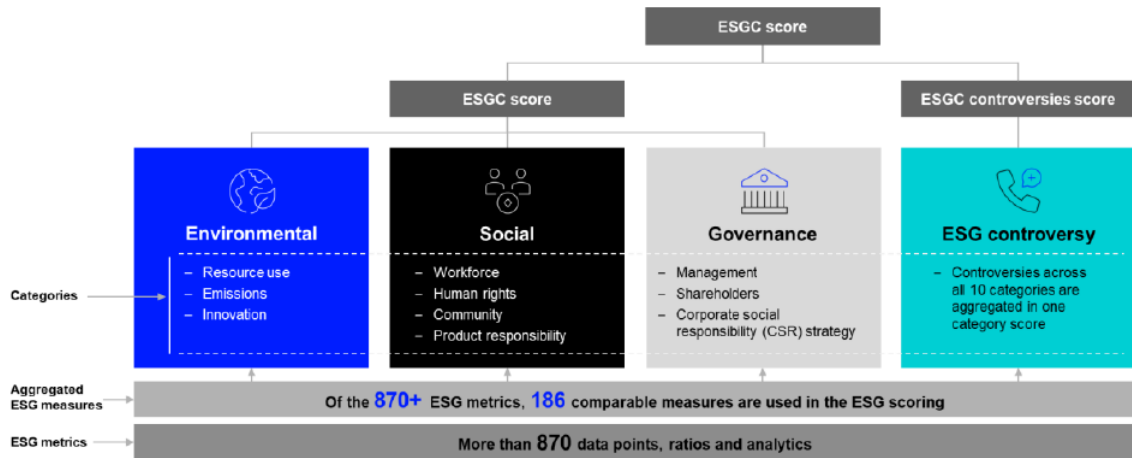
LSEG-Refinitiv collects **publicly available information** from various sources (company disclosures, but also media reports)

This is different from other data providers (e.g., MSCI) that survey firms

Getting from the underlying data points to the final score is complex: it involves **combining different data types** (numeric and Boolean) and (industry-specific) weightings

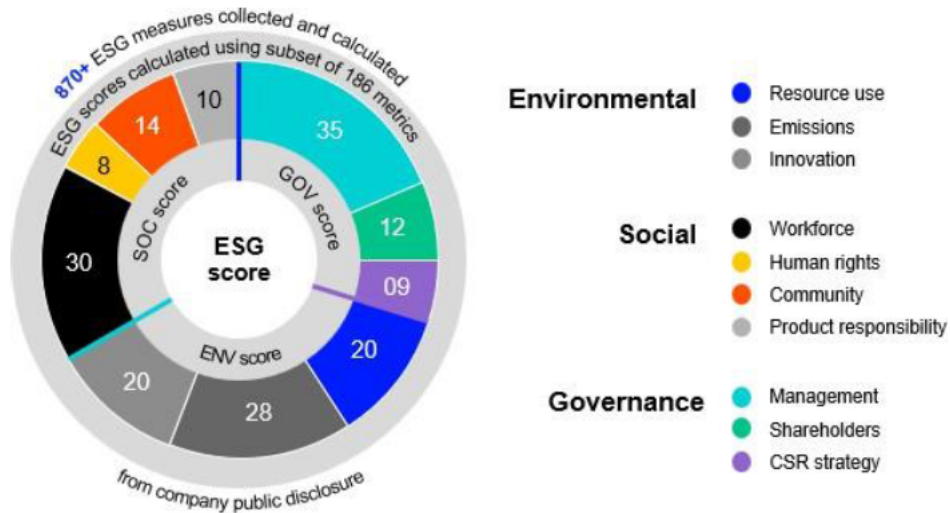
This is not specific to the Refinitiv ESG data

LSEG-Refinitiv-ESG Scoring Methodology



Source: LSEG (2024), *Environmental, Social and Governance (ESG) Scores from LSEG*

LSEG-Refinitiv-ESG



Source: LSEG (2024), *Environmental, Social and Governance (ESG) Scores from LSEG*

Scoring System – Step 1: ESG Category Scores

Boolean data (“Yes”, “No”, “Null”)

- Positive polarity: Yes= 1, No/Null= 0
- Negative polarity: Yes/Null= 0, No= 1

Numeric data: Relative percentile ranking

For each indicator k , in a given industry, we compute the score of firm i as follows:

$$\text{Score}_{i,k} = \frac{\#\{j : X_{j,k} < X_{i,k}\} + \frac{1}{2}\#\{j : X_{j,k} = X_{i,k}\}}{\# \text{ firms}}.$$

Example 1: There are 12 firms in the industry. For the question “Does the company have a policy to improve emission reduction?” (positive polarity), 5 firms have such a policy, 3 firms do not, and 4 do not disclose this information. Therefore, the score for the 5 firms with a policy is

$$(7 + 5/2)/12 = 0.7917.$$

All the other firms obtain a score equal to 0.

Scoring System – Step 1

Example 2: For numerical indicator, there is in general no tie between firms. So, the formula is usually equivalent to the relative rank.

Remark: the normalization (ranking) is done at the sector level, not at the universe level.

ESG scores are usually relative (w.r.t. a sector), not absolute

Best-in-class or **Worst-in-class** logic

The score of firm i for the category c is the sum of the scores for all indicators in the category:

$$\text{Score}_{i,c} = \sum_{k=1}^{K_c} \text{Score}_{i,k},$$

where K_c is the number of indicators for category c .

Scoring System – Step 2: Materiality Matrix

For each theme within a category, we measure if this theme is material for a given industry. This provides the category weights to be used in computing scores.

Two ways to compute the materiality matrix:

- **Industry median:** for numeric indicator (say, carbon emissions), the materiality is based on the rank of a given industry for the median carbon emissions in the industry
- **Transparency weight:** for Boolean indicators, the materiality is based on the level of disclosure in a given industry (proportion of “Yes”)

Example: Category weights for some industries ($w_{j,E} + w_{j,S} + w_{j,G} = 1$)

Industry _j	Emission	Innovation	Resource use	E pillar ($w_{j,E}$)	S pillar ($w_{j,S}$)	G pillar ($w_{j,G}$)
Banking services	0.02	0.10	0.02	0.14	0.50	0.36
Coal	0.20	0.02	0.19	0.41	0.28	0.30
Construction materials	0.15	0.12	0.15	0.42	0.34	0.24
Oil and gas	0.11	0.10	0.13	0.34	0.42	0.24
Water and related utilities	0.15	0.13	0.15	0.43	0.31	0.25

Scoring System – Step 3: Pillar score and ESG score

The pillar score (say, E score) of a firm i in industry j is given by:

$$\text{Score}_i^{(E)} = \sum_{c=1}^{N_E} w_{j,c} \text{Score}_{i,c},$$

where $w_{j,c}$ is the relative weight of the score of category c for industry j (e.g., 0.15/0.43 for Emission in Water industry) and $\text{Score}_{i,c}$ is the score of category c for firm i .

The overall ESG score of a firm i in industry j is given by:

$$\text{Score}_i^{(ESG)} = w_{j,E} \text{Score}_i^{(E)} + w_{j,S} \text{Score}_i^{(S)} + w_{j,G} \text{Score}_i^{(G)},$$

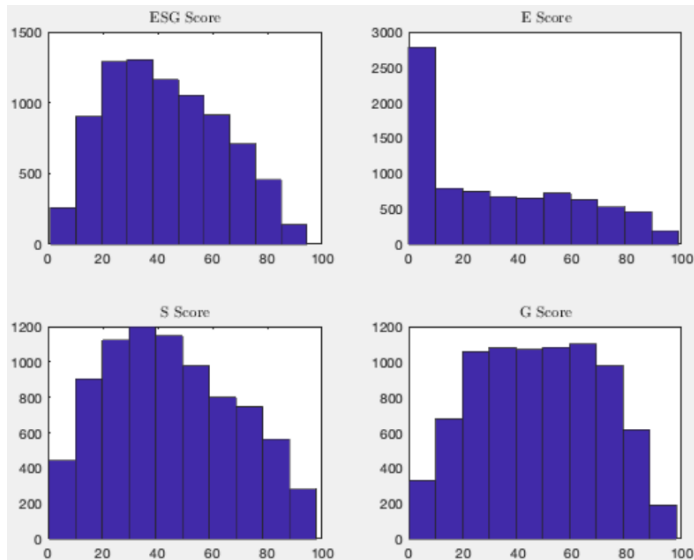
where $w_{j,E}$ is the relative weight of the E pillar for industry j (e.g., 0.43 for E pillar in Water industry) and $\text{Score}_i^{(E)}$ is the E pillar score for firm i .

Scoring System – Step 4: Ratings

Score range	Grade	Description
0.0 <= score <= 0.083333	D -	'D' score indicates poor relative ESG performance and insufficient degree of transparency in reporting material ESG data publicly.
0.083333 < score <= 0.166666	D	
0.166666 < score <= 0.250000	D +	
0.250000 < score <= 0.333333	C -	'C' score indicates satisfactory relative ESG performance and moderate degree of transparency in reporting material ESG data publicly.
0.333333 < score <= 0.416666	C	
0.416666 < score <= 0.500000	C +	
0.500000 < score <= 0.583333	B -	'B' score indicates good relative ESG performance and above- average degree of transparency in reporting material ESG data publicly.
0.583333 < score <= 0.666666	B	
0.666666 < score <= 0.750000	B +	
0.750000 < score <= 0.833333	A -	'A' score indicates excellent relative ESG performance and high degree of transparency in reporting material ESG data publicly.
0.833333 < score <= 0.916666	A	
0.916666 < score <= 1	A +	

Source: LSEG (2024), *Environmental, Social and Governance (ESG) Scores from LSEG*

Distribution of Scores with LSEG-Refinitiv ESG (end of 2019)



Diagnosis by IOSCO (International Organization of Securities Commissions) (2021)

- Little clarity and alignment on definitions, on what ratings intend to measure
- Lack of transparency about the methodologies underpinning the ratings
- Wide divergence within the ESG rating
- Uneven coverage of products offered, with certain industries or geographical areas benefiting from more coverage than others, thereby leading to gaps for investors seeking to follow certain investment strategies
- Potential conflicts of interest where the ESG ratings provider (or an entity closely associated with the provider) performs consulting services for companies that are the subject of these ESG ratings
- Lack of communication with companies that are the subject of ESG ratings

Issues with ESG Scores

Berg, Koelbel, Rigobon (2019), "Aggregate Confusion: The Divergence of ESG Ratings"

Comparison of scores provided by several data providers:

- SA (Sustainalytics), VI (Vigeo-Eiris), KL (MSCI-KLD), RS (RobecoSAM), and A4 (Refinitiv Asset4)
- Correlation between ratings at aggregate ESG level and at E, S, and G level

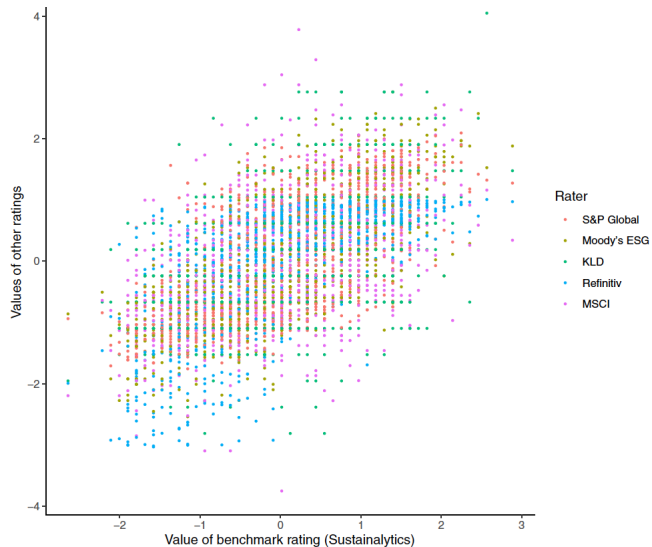
	KL SA	KL MO	KL SP	KL RE	KL MS	SA MO	SA SP	SA RE	SA MS	MO SP	MO RE	MO MS	SP RE	SP MS	RE MS	Average
ESG	0.53	0.49	0.44	0.42	0.53	0.71	0.67	0.67	0.46	0.7	0.69	0.42	0.62	0.38	0.38	0.54
E	0.59	0.55	0.54	0.54	0.37	0.68	0.66	0.64	0.37	0.73	0.66	0.35	0.7	0.29	0.23	0.53
S	0.31	0.33	0.21	0.22	0.41	0.58	0.55	0.55	0.27	0.68	0.66	0.28	0.65	0.26	0.27	0.42
G	0.02	0.01	-0.01	-0.05	0.16	0.54	0.51	0.49	0.16	0.76	0.76	0.14	0.79	0.11	0.07	0.30

Average correlation between aggregate ESG scores: 0.61

Average correlation between all E, S, and G scores: 0.52

Remark: Correlation between credit ratings from Moody's and Standard & Poor's: 0.986

ESG rating disagreement (Berg, Koelbel, Rigobon, 2019)



Source of Disagreement

Disagreement is decomposed into three sources:

- **53%** is due to **measurement**: raters measure the same ESG attribute with different indicators. For example:
 - Labor practices could be evaluated based on workforce turnover, or number of labor cases against the firm
 - Female friendliness could be measured by the gender pay gap, the percentage of women on the board, or the percentage of women in the workforce
- **44%** is due to **scope**: different raters include different attributes
 - Most raters consider a firm's greenhouse gas emissions when evaluating its environmental record, but only some will include electromagnetic radiation
 - One rating agency may include lobbying, another might not
- **3%** is due to **weight**: different raters place different weights on the individual components when calculating the overall score

Sovereign ESG Data

Sovereign data can be used for evaluating a portfolio of sovereign bonds.

World Bank's framework (number of indicators)

Environment (27)	Social (22)	Governance (18)
Emissions & pollution (5)	Education & skills (3)	Human rights (2)
Natural capital endowment & mgt (6)	Employment (3)	Government effectiveness (2)
Energy use & security (7)	Demography (3)	Stability & rule of law (4)
Environment/climate risk & resilience (6)	Poverty & inequality (4)	Economic environment (3)
Food security (3)	Health & nutrition (5)	Gender (4)
	Access to services (4)	Innovation (3)

Source: <https://datatopics.worldbank.org/esg/methodology.html> and
<https://datatopics.worldbank.org/esg/framework.html>

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ESG Investing in World Equity

Alessandrini and Jondeau (2020), “ESG Investing: From Sin Stocks to Smart Beta”

Period: January 2007–December 2018

Universe: World equity

- MSCI-ESG universe
- Approx. 1'700 firms in 2007
- Approx. 7'000 firms in 2018

We restrict the universe to the MSCI-ACWI benchmark (approx. 2'700 firms) that covers large and mid-cap stocks across 23 developed and 24 emerging markets

Scores:

- Between 0 and 10, 10 is the best
- Individual E/S/G scores
- Score adjusted for the industry

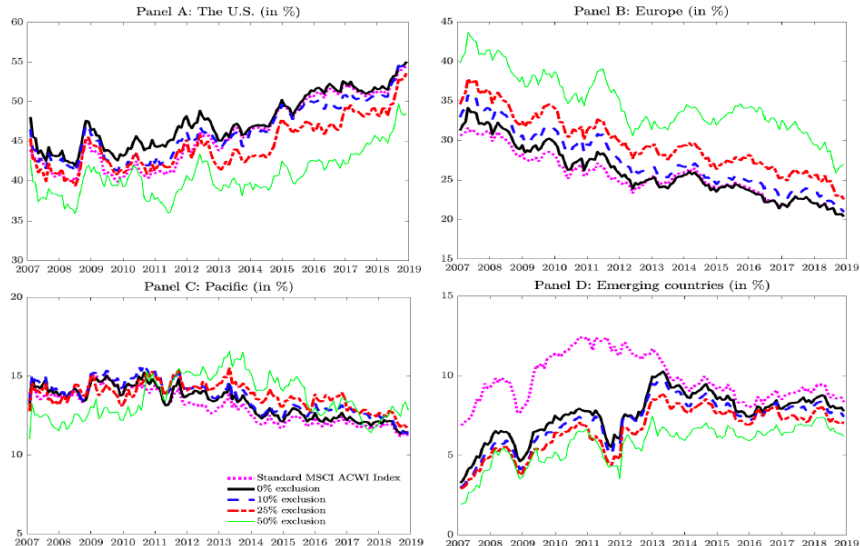
We consider several alternative specifications (different scores; different filters: firms with 10%, 25%, 50% worst scores are excluded)

ESG Exclusion in World Equity

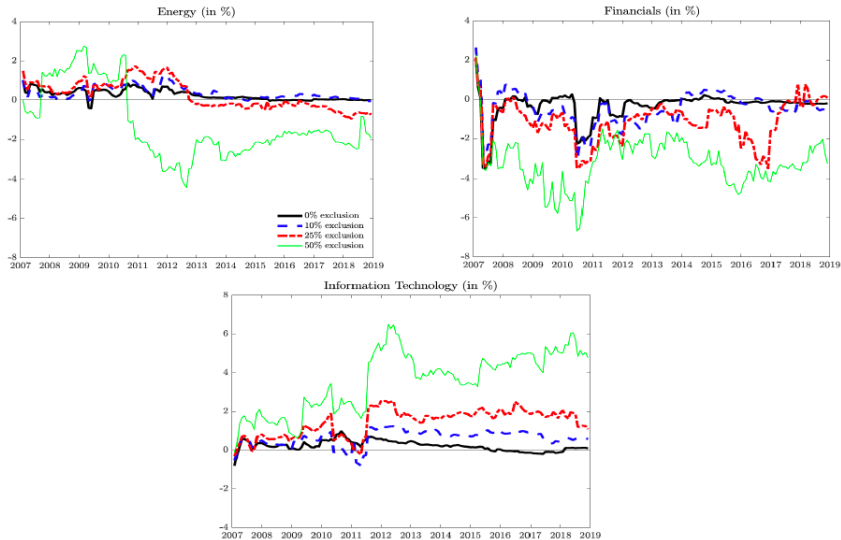
	Average ESG				Environment			
	0% excluded	10% excluded	25% excluded	50% excluded	0% excluded	10% excluded	25% excluded	50% excluded
Average score	5.57	5.92	6.36	7.16	5.83	6.07	6.39	6.98
Annual return	5.74	5.54	5.60	5.63	5.74	5.70	5.88	6.33
Annual volat.	15.70	15.67	15.61	15.62	15.70	15.68	15.67	15.51
Sharpe ratio	0.32	0.31	0.31	0.31	0.32	0.32	0.33	0.36

	Social				Governance			
	0% excluded	10% excluded	25% excluded	50% excluded	0% excluded	10% excluded	25% excluded	50% excluded
Average score	5.06	5.32	5.66	6.23	5.61	5.93	6.27	7.01
Annual return	5.74	5.59	5.48	5.51	5.74	5.44	5.40	5.62
Annual volat.	15.70	15.69	15.73	15.87	15.70	15.63	15.56	15.77
Sharpe ratio	0.32	0.31	0.30	0.30	0.32	0.30	0.30	0.31
Tracking error	0.64	0.74	0.93	1.42	0.68	0.93	1.11	1.42

Regional Exposures – Average ESG Score



Sectoral Exposures – Average ESG Score



ESG Investing in World Bonds

Alessandrini, Baptista Balula, and Jondeau (2022), “ESG Screening in the Fixed-Income Universe”

Period: January 2014–December 2020

Universe: World bonds

- Bloomberg Barclays Global Aggregate Corporate Index
- Approx. 6'405 securities in 2014 (817 firms)
- Approx. 11'117 securities in 2020 (1'121 firms)

Matching with MSCI-ESG data

Several alternative specifications

- Different currency segments (USD, EUR)
- Different scores (ESG, E, S, G)
- Different filters (firms with 10%, 25%, 50% worst scores are excluded)

ESG Exclusion in World Bonds – USD Segment

	Average ESG				Environment			
	0% excluded	25% excluded	33% excluded	50% excluded	0% excluded	25% excluded	33% excluded	50% excluded
Average Score	4.97	5.83	6.10	6.69	6.16	7.12	7.38	7.94
Annual return	4.31	4.33	4.34	4.36	4.31	4.35	4.36	4.34
Annual volat.	5.22	5.20	5.09	5.07	5.22	4.91	4.88	4.68
Sharpe ratio	0.66	0.67	0.69	0.69	0.66	0.71	0.72	0.75
Tracking error	0.09	0.24	0.35	0.51	0.09	0.60	0.66	0.84

	Social				Governance			
	0% excluded	25% excluded	33% excluded	50% excluded	0% excluded	25% excluded	33% excluded	50% excluded
Average Score	4.45	5.04	5.21	5.58	4.89	5.57	5.74	6.10
Annual return	4.31	4.34	4.35	4.33	4.31	4.38	4.41	4.39
Annual volat.	5.22	5.24	5.24	5.24	5.22	5.47	5.52	5.61
Sharpe ratio	0.66	0.67	0.67	0.67	0.66	0.65	0.64	0.63
Tracking error	0.09	0.18	0.19	0.23	0.09	0.29	0.33	0.46

Strategy:

- ❶ Exclude firms with the lowest ESG score corresponding to 25% of the market value of the portfolio in a given region-sector
- ❷ Reinvest the proceeds in firms with the highest ESG score in the same region-sector

This approach has two advantages:

- It overweighs firms with high scores relative to firms with low scores. In the short term, it reduces the cost of financing of high-score firms
- It maintains the region-sector risk exposures of the initial portfolio

Potential drawback:

- It may not maximize the increase in score relative to exclusion strategy

ESG Best-In-Class in World Bonds – USD Segment

	Average ESG				Environment			
	0% excluded	25% excluded	33% excluded	50% excluded	0% excluded	25% excluded	33% excluded	50% excluded
Average Score	4.97	6.07	6.24	6.39	6.16	7.17	7.31	7.43
Annual return	4.31	4.34	4.35	4.35	4.31	4.29	4.27	4.28
Annual volat.	5.22	5.13	5.16	5.06	5.22	5.12	5.08	5.06
Sharpe ratio	0.66	0.68	0.68	0.69	0.66	0.67	0.67	0.68
Tracking error	0.09	0.28	0.25	0.33	0.09	0.31	0.37	0.45

	Social				Governance			
	0% excluded	25% excluded	33% excluded	50% excluded	0% excluded	25% excluded	33% excluded	50% excluded
Average Score	4.45	5.19	5.30	5.38	4.89	5.63	5.77	5.90
Annual return	4.31	4.34	4.35	4.35	4.31	4.35	4.36	4.36
Annual volat.	5.22	5.18	5.17	5.15	5.22	5.25	5.31	5.30
Sharpe ratio	0.66	0.67	0.68	0.68	0.66	0.67	0.66	0.66
Tracking error	0.09	0.19	0.22	0.28	0.09	0.10	0.14	0.16

Deconstructing ESG Scores

Ehlers, Elsenhuber, Jegarasasingam and Jondeau (2023), *Deconstructing ESG scores: How to invest with your own criteria?*

Two issues raised by disclosure of ESG data

- **Missing values in numeric indicators**

- What if a firm does not report on a numeric indicator?
- Refinitiv reports a missing value
- Proportion of valid values is relatively low ($\sim 40\%$ on average)

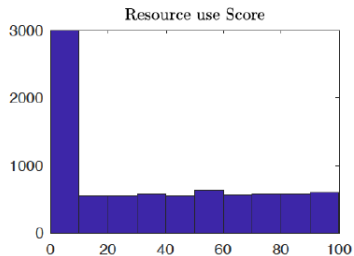
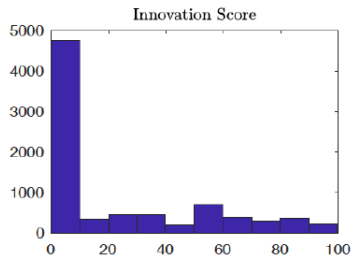
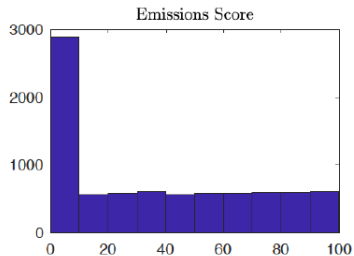
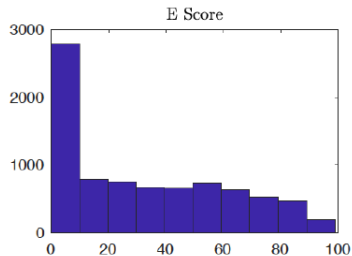
- **Proportion of zeros in Boolean indicators**

- What if a firm does not report on a Boolean indicator?
- Refinitiv reports a value of 0 (negative answer) to incentivize companies to improve their information disclosure
- Proportion of zeros is large at the category level (e.g., 74% of zeros for the reporting on firm's environmental expenditures)

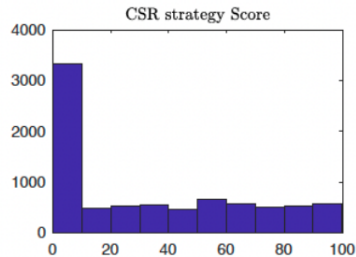
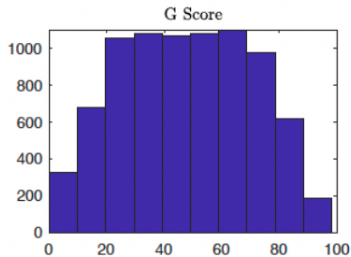
Proportion of Valid and Zero Scores

Category	Numeric items	Proportion of valid values			Proportion of zeros		
		2010	2015	2019	2010	2015	2019
ESG	63	37.2	39.1	39.7	0	0	0
Environment	27	18.9	19.6	20.8	17.9	21.0	21.2
Emission reduction	14	21.8	22.4	21.3	26.6	29.8	28.2
Innovation	7	12.1	11.1	16.1	56.1	56.6	56.5
Resource use	6	20.3	23.2	24.8	27.6	29.1	29.4
Social	19	24.1	29.2	30.5	0	0	0
Community	1	39.3	37.0	35.2	0.6	0	0
Human rights	0	–	–	–	66.6	55.4	41.9
Product responsibility	2	4.9	13.1	12.9	24.8	17.2	10.0
Workforce	16	25.5	30.7	32.4	0	0	0
Governance	17	80.9	81.2	80.1	0	0	0
CSR strategy	0	–	–	–	35.1	36.4	33.9
Management	14	80.8	80.0	78.6	0	0	0
Shareholders	3	81.4	86.9	87.3	0	0	0

Distribution of E Scores and Underlying Categories (2019)



Distribution of G Scores and Underlying Categories (2019)



Portfolio Allocation Exercise: Global Screening

Screening strategy

- Exclude 10%, 25%, 33% of firms with the lowest scores (in terms of market value)
- Reinvest the proceeds in all remaining firms in proportion of market weights

We cannot apply the 10% strategy for some categories

- with a large proportion of zero scores:
- Innovation (40% of zeros), Human rights (40% of zeros), CSR strategy (20% of zeros)

33% strategy gives interesting results:

- Higher score gain for categories (+18pp instead of 14pp for pillars, 11pp for ESG)
- No reduction in financial performance (same Sharpe ratio, moderate tracking error)
- But regional and sectoral imbalances

Summary Statistics: Global Screening (33% Threshold)

Category	Prop. firms zeros	Prop. mkt cap zeros	Prop. firms exclud.	Prop. mkt cap exclud.	Bench-mark score	Port-folio score	Score gain	Ann. return	Ann. vola-tility	Sharpe ratio
MSCI ACWI Benchmark	–	–	0.0	0.0	–	–	–	9.71	14.07	0.65
	–	–	5.2	2.4	–	–	–	9.91	14.34	0.65
ESG	0.0	0.0	60.2	33.0	64.0	75.2	11.2	9.70	14.29	0.64
Environment	8.9	4.1	55.7	32.6	61.2	77.1	15.9	8.99	14.16	0.60
Emission	13.6	6.2	55.2	32.9	66.9	84.4	17.5	9.33	14.51	0.61
Innovation	40.3	26.1	49.5	33.5	44.1	64.8	20.7	10.12	14.36	0.67
Resource Use	13.9	6.0	59.1	32.9	69.2	86.9	17.7	9.64	14.42	0.63
Social	0.0	0.0	58.4	32.9	65.4	78.1	12.7	9.62	14.42	0.63
Community	0.6	0.2	57.4	32.8	73.4	88.4	15.0	10.53	14.45	0.69
Human Rights	39.8	21.9	55.0	33.1	50.3	71.7	21.4	9.76	14.44	0.64
Product Resp.	13.0	4.5	48.2	33.1	61.0	79.2	18.2	9.56	14.12	0.64
Workforce	0.0	0.0	53.1	32.9	72.8	86.8	14.1	9.45	14.51	0.61
Governance	0.0	0.0	49.6	32.9	63.7	76.4	12.7	9.97	14.40	0.65
CSR Strategy	18.7	9.5	52.8	33.4	61.9	81.6	19.7	8.99	14.36	0.59
Management	0.0	0.0	46.7	32.8	65.9	82.0	16.1	10.35	14.42	0.68
Shareholders	0.0	0.0	40.0	33.0	57.4	74.4	17.0	9.95	14.32	0.66

End of Lecture 10